



C#

Level 1

Bitwise and Bit Shift Operators



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Definition

Bitwise Operators

 Work directly with the individual bits (0 and 1) of integer values.

Bitwise Operators

 $\&$ $\rightarrow$ Bitwise AND

 $|$ $\rightarrow$ Bitwise OR

 $\wedge$ $\rightarrow$ Bitwise XOR

 $\sim$ $\rightarrow$ Bitwise NOT

Bit Shift Operators

 $\ll$ $\rightarrow$ Left Shift

 $\gg$ $\rightarrow$ Right Shift

 $\ggg$ $\rightarrow$ Unsigned Right Shift

Bitwise Operations

AND &

 Result bit is 1 only when both bits are 1.

OR |

 Result bit is 1 when at least one bit is 1.

XOR ^

 Result bit is 1 when the bits are different.

NOT ~

 Reverses every bit: $0 \rightarrow 1$ and $1 \rightarrow 0$.

 For signed integers, the result uses two's-complement representation.

Bit Shift Operations

Left Shift <<

-  Moves bits to the left.

-  For suitable values, each shift left by 1 corresponds to multiplying by 2.

-  $5 \ll 1 \rightarrow 10$

Right Shift >>

-  Moves bits to the right.

-  For positive integers, each shift right by 1 corresponds to integer division by 2.

-  $20 \gg 1 \rightarrow 10$

Unsigned Right Shift >>>

-  Moves bits right and fills the left side with zeros.

-  Does not preserve the sign bit.

💡 Example

5 = 0101

3 = 0011

5 & 3 → 0001 → 1

5 | 3 → 0111 → 7

5 ^ 3 → 0110 → 6

Note

- ◆ & and && are different operators.
- ◆ | and || are different operators.
- ◆ Bitwise operators are commonly used with flags, masks, binary data, and low-level programming.

Comparison

 `&` → Bitwise AND → works on bits

 `|` → Bitwise OR → works on bits

 `^` → Bitwise XOR → works on bits

 `~` → Bitwise NOT → works on bits

 `&&`

✓ Logical AND → works with Boolean conditions.

 `||`

✓ Logical OR → works with Boolean conditions.

 `!`

✓ Logical NOT → works with Boolean values.

✦ Summary

- ✓ Bitwise operators work with individual bits.
- ✓ $\&$ $\rightarrow$ AND
- ✓ $|$ $\rightarrow$ OR
- ✓ $\wedge$ $\rightarrow$ XOR
- ✓ $\sim$ $\rightarrow$ NOT
- ✓ $\ll$ $\rightarrow$ Left Shift
- ✓ $\gg$ $\rightarrow$ Right Shift
- ✓ $\ggg$ $\rightarrow$ Unsigned Right Shift

Thank you 😊